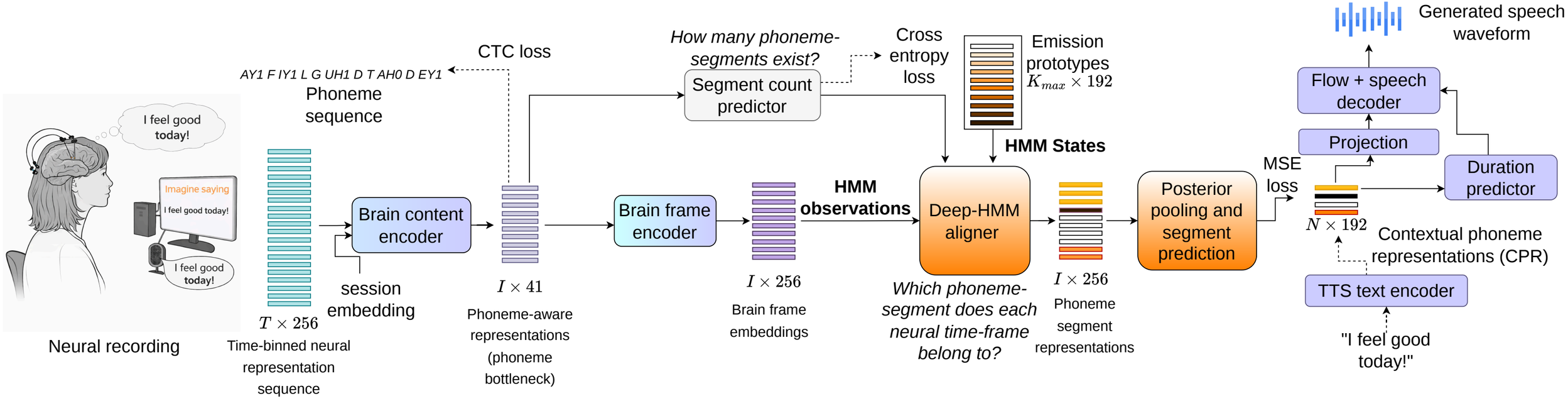
I feel good today!
Imagine saying
I feel good today!
I feel good today!
Neural recording
AY1 F IY1 L G UH1 D T AH0 D EY1
Phoneme sequence
CTC loss
$T \times 256$
Time-binned neural representation sequence
Brain content encoder
session embedding
$I \times 41$
Phoneme-aware representations (phoneme bottleneck)
How many phoneme-segments exist?
Segment count predictor
Cross entropy loss
Brain frame encoder
$I \times 256$
Brain frame embeddings
HMM observations
Emission prototypes $K_{max} \times 192$
HMM States
Deep-HMM aligner
Which phoneme-segment does each neural time-frame belong to?
$I \times 256$
Phoneme segment representations
Posterior pooling and segment prediction
MSE loss
$N \times 192$
Contextual phoneme representations (CPR)
Projection
Duration predictor
Flow + speech decoder
Generated speech waveform
TTS text encoder
"I feel good today!"